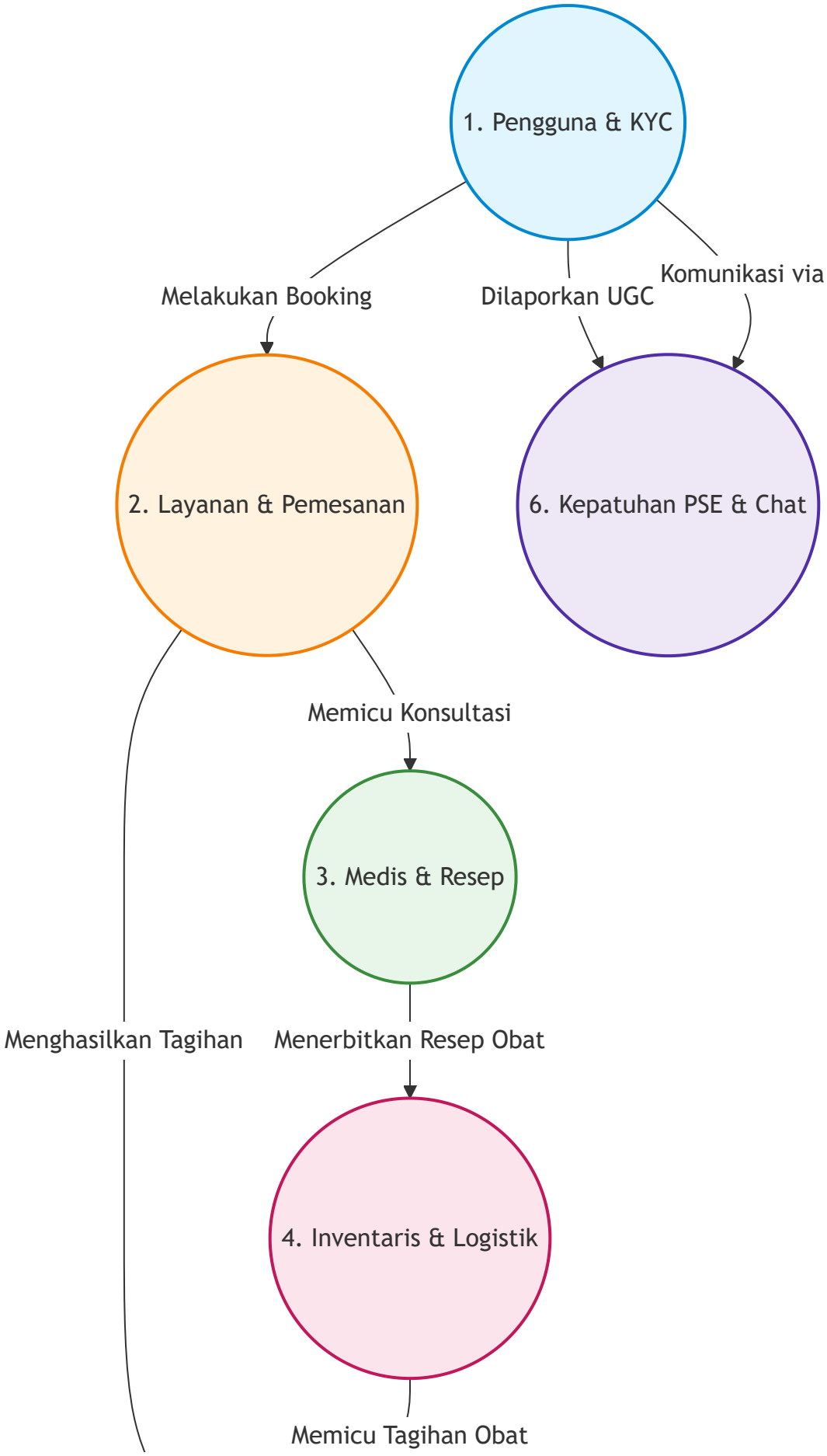
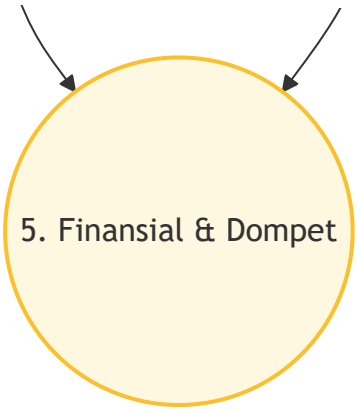
6. Database Schema (Entity Relationship Diagram)

Dokumen ini mendefinisikan struktur penyimpanan data (*database*) untuk sistem Telemedicine berbasis *Relational Database* (MySQL/PostgreSQL). Karena sistem ini menaungi proses yang kompleks, struktur tabel dikelompokkan menjadi 6 sub-sistem/domain yang saling berinteraksi guna memudahkan pemahaman desain (*Maintainability*).

6.1. Diagram Korelasi Antar Sub-sistem (High-Level)

Diagram ini memberikan gambaran besar bagaimana ke-6 domain basis data terhubung dan berinteraksi dalam satu kesatuan sistem.


1. Pengguna & KYC
Melakukan Booking
Dilaporkan UGC
Komunikasi via
2. Layanan & Pemesanan
6. Kepatuhan PSE & Chat
Memicu Konsultasi
3. Medis & Resep
Menghasilkan Tagihan
Menerbitkan Resep Obat
4. Inventaris & Logistik
Memicu Tagihan Obat



6.2. ERD Domain 1: Pengguna, Profil & Legalitas (KYC)

Sub-sistem ini mengelola autentikasi pengguna beserta detail profil medis. Termasuk pula penyimpanan dokumen legalitas (KYC) milik Dokter dan Apotek yang diperlukan untuk verifikasi operasional.

users			
bigint	id	PK	
string	name		
string	email	UK	
string	whatsapp_number	UK	
string	password		
enum	role		patient, doctor, pharmacy, admin, superadmin
enum	status		active, pending, suspended
timestamp	created_at		

memiliki (has)

memiliki (has)

patient_profiles			
bigint	id	PK	
bigint	user_id	FK	
date	dob		Tanggal Lahir
enum	gender		M, F
text	address		
string	emergency_contact		
enum	blood_type		A, B, AB, O

partner_profiles			
bigint	id	PK	
bigint	user_id	FK	
string	str_number		Nomor STR Dokter
string	sip_number		Nomor SIP Dokter
string	sia_number		Nomor SIA Apotek
string	sipa_number		Nomor SIPA Apoteker
string	specialization		
string	kyc_document_url		
enum	kyc_status		pending, approved, rejected

6.3. ERD Domain 2: Katalog, Jadwal & Skrining (Anamnesis)

Sub-sistem yang menyimpan master data layanan penurunan berat badan, ketersediaan jadwal dokter, proses reservasi (*booking*), serta kuesioner medis pasien (Anamnesis).

service_packages		
bigint	id	PK
string	name	
text	description	
decimal	price_consultation	
decimal	price_medicine	
boolean	is_active	

doctor_schedules			
bigint	id	PK	
bigint	doctor_id	FK	users.id
date	schedule_date		
time	start_time		
time	end_time		
boolean	is_booked		

dipesan dalam (ordered in)

direservasi oleh (reserved by)

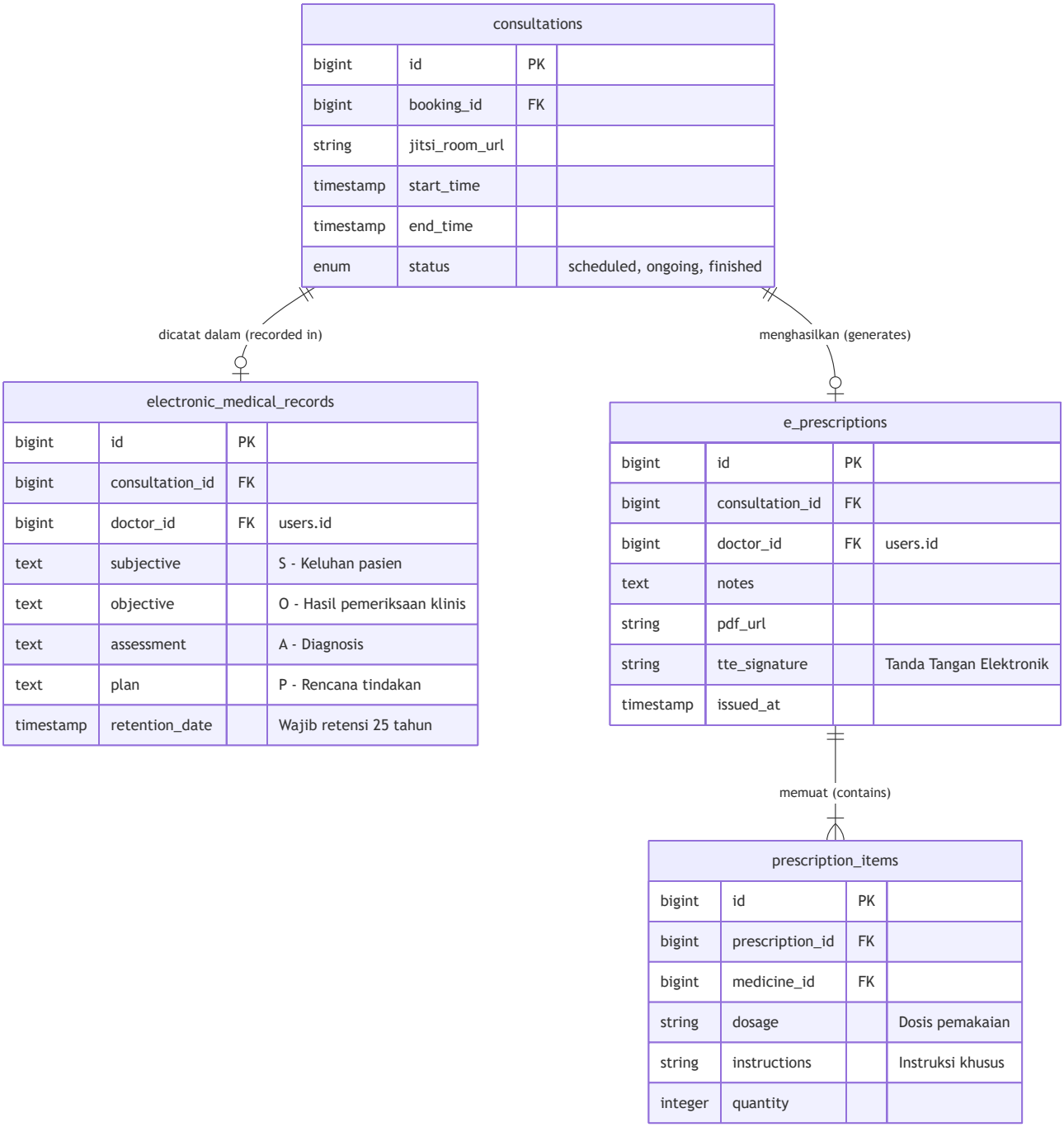
bookings			
bigint	id	PK	
bigint	patient_id	FK	users.id
bigint	doctor_schedule_id	FK	
bigint	service_package_id	FK	
enum	status		pending, paid, cancelled, completed
timestamp	created_at		

mewajibkan (requires)

anamnesis_records			
bigint	id	PK	
bigint	booking_id	FK	
bigint	patient_id	FK	users.id
json	questions_answers		Format kuesioner dinamis
enum	screening_status		passed, warning, blocked

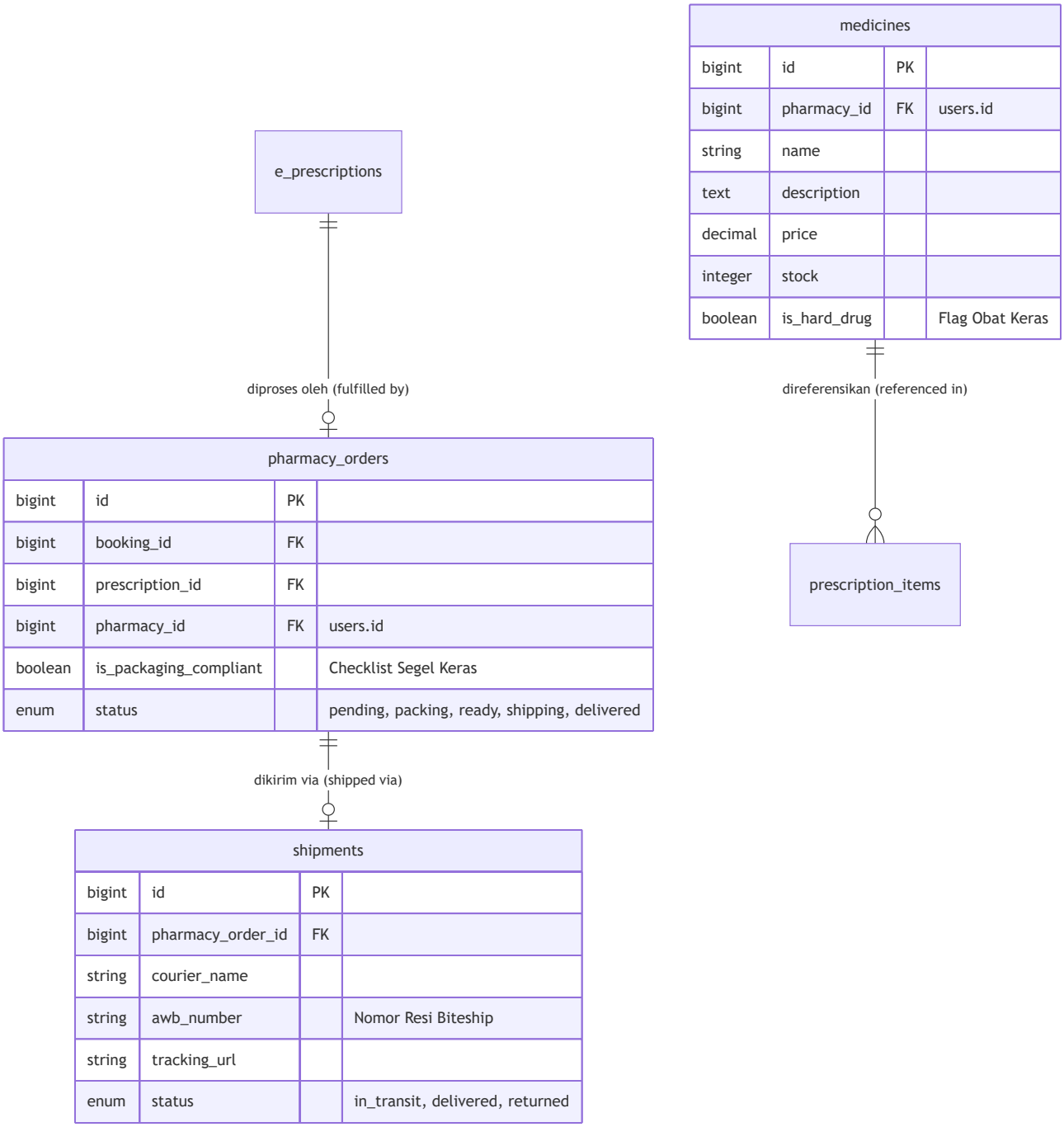
6.4. ERD Domain 3: Rekam Medis Elektronik (RME) & Resep Digital

Sub-sistem klinis yang menangani sesi ruang virtual, form RME/SOAP, serta pembuatan entitas Resep Elektronik (*E-Prescription*) ber-TTE.


consultations
bigint id PK
bigint booking_id FK
string jitsi_room_url
timestamp start_time
timestamp end_time
enum status scheduled, ongoing, finished
dicatat dalam (recorded in)
menghasilkan (generates)
electronic_medical_records
bigint id PK
bigint consultation_id FK
bigint doctor_id FK users.id
text subjective S - Keluhan pasien
text objective O - Hasil pemeriksaan klinis
text assessment A - Diagnosis
text plan P - Rencana tindakan
timestamp retention_date Wajib retensi 25 tahun
e_prescriptions
bigint id PK
bigint consultation_id FK
bigint doctor_id FK users.id
text notes
string pdf_url
string tte_signature Tanda Tangan Elektronik
timestamp issued_at
memuat (contains)
prescription_items
bigint id PK
bigint prescription_id FK
bigint medicine_id FK
string dosage Dosis pemakaian
string instructions Instruksi khusus
integer quantity

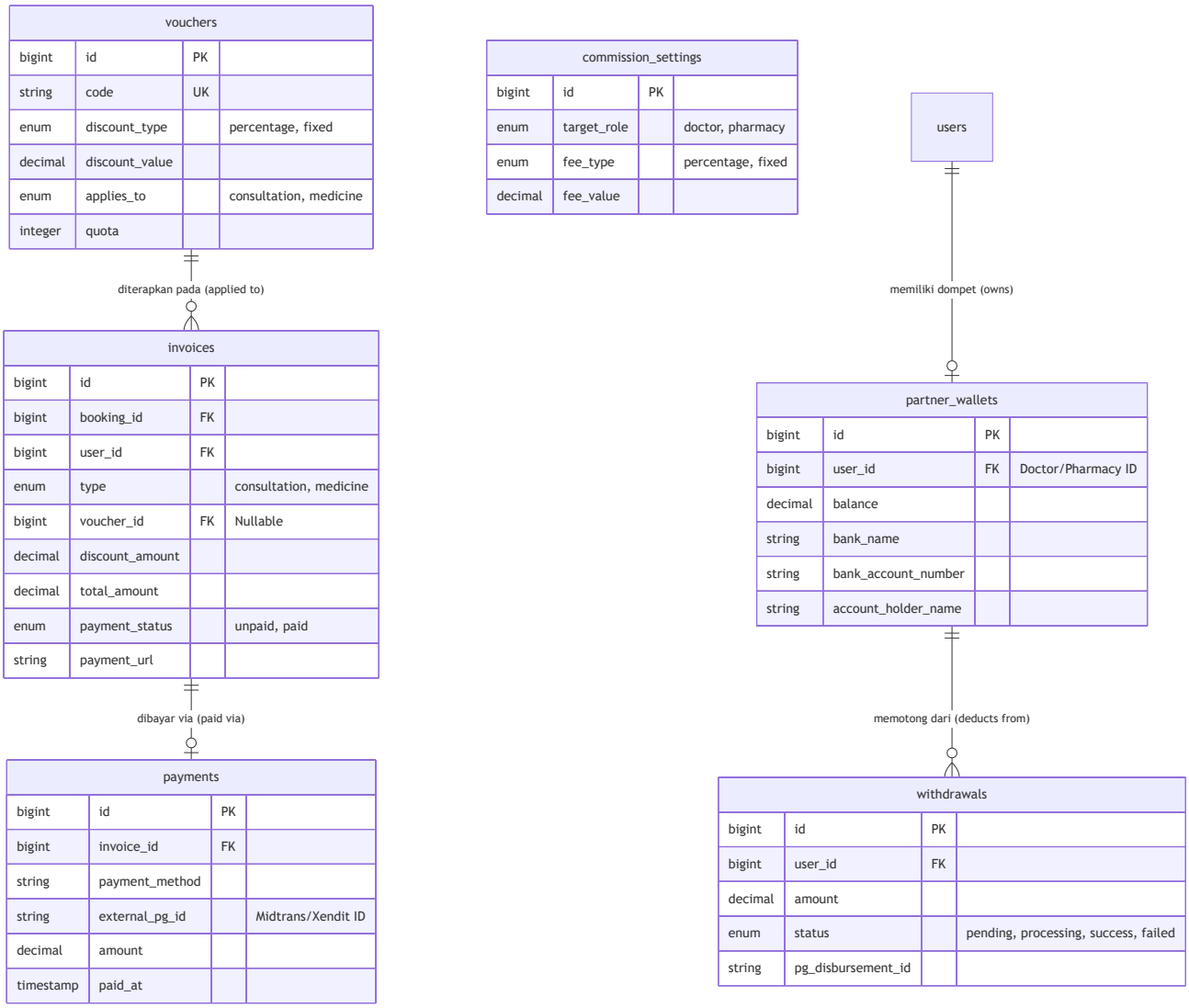
6.5. ERD Domain 4: Inventaris & Logistik (Fulfillment)

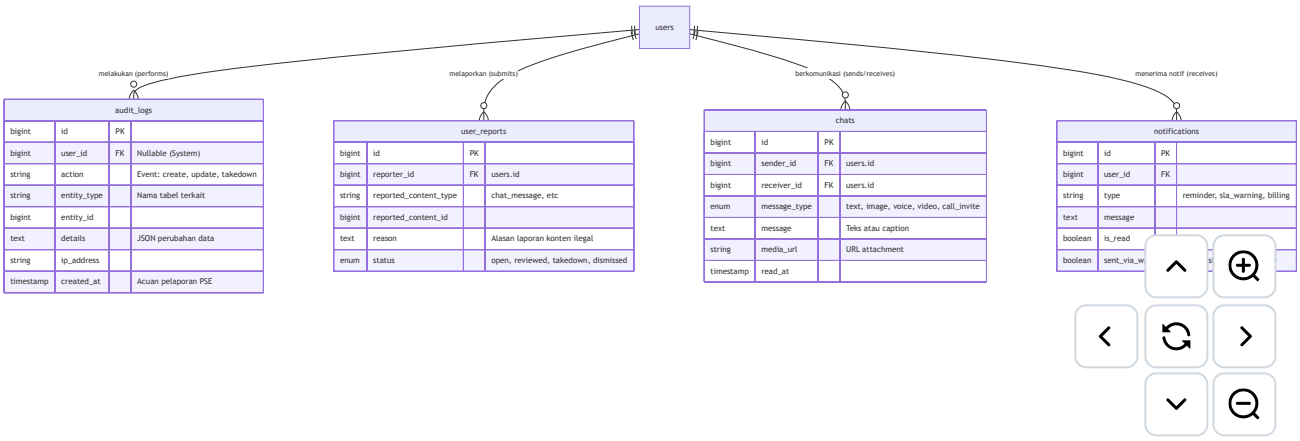
Sub-sistem pengelolaan ketersediaan obat (Farmasi/Apotek) fisik, manajemen pengemasan (*checklist* regulasi obat keras), hingga pelacakan pengiriman kurir.


medicines
bigint id PK
bigint pharmacy_id FK users.id
string name
text description
decimal price
integer stock
boolean is_hard_drug Flag Obat Keras
direferensikan (referenced in)
prescription_items
e_prescriptions
diproses oleh (fulfilled by)
pharmacy_orders
bigint id PK
bigint booking_id FK
bigint prescription_id FK
bigint pharmacy_id FK users.id
boolean is_packaging_compliant Checklist Segel Keras
enum status pending, packing, ready, shipping, delivered
dikirim via (shipped via)
shipments
bigint id PK
bigint pharmacy_order_id FK
string courier_name
string awb_number Nomor Resi Biteship
string tracking_url
enum status in_transit, delivered, returned

6.6. ERD Domain 5: Finansial (Billing, Voucher, Payout)

Pusat perputaran uang. Mengelola tagihan berantai (*Double Billing*), integrasi pembayaran (Webhook), voucher promosi, kalkulasi rasio komisi platform, hingga riwayat dompet dan penarikan tunai mitra.


vouchers
bigint id PK
string code UK
enum discount_type percentage, fixed
decimal discount_value
enum applies_to consultation, medicine
integer quota
diterapkan pada (applied to)
invoices
bigint id PK
bigint booking_id FK
bigint user_id FK
enum type consultation, medicine
bigint voucher_id FK Nullable
decimal discount_amount
decimal total_amount
enum payment_status unpaid, paid
string payment_url
dibayar via (paid via)
payments
bigint id PK
bigint invoice_id FK
string payment_method
string external_pg_id Midtrans/Xendit ID
decimal amount
timestamp paid_at
commission_settings
bigint id PK
enum target_role doctor, pharmacy
enum fee_type percentage, fixed
decimal fee_value
users
memiliki dompet (owns)
partner_wallets
bigint id PK
bigint user_id FK Doctor/Pharmacy ID
decimal balance
string bank_name
string bank_account_number
string account_holder_name
memotong dari (deducts from)
withdrawals
bigint id PK
bigint user_id FK
decimal amount
enum status pending, processing, success, failed
string pg_disbursement_id

6.7. ERD Domain 6: Kepatuhan PSE, SLA & Komunikasi

Sub-sistem pencatatan rekam jejak aktivitas sistem (*Audit Trail*) untuk pemenuhan permintaan Aparat Hukum, pelaporan konten ilegal (UGC), serta fungsionalitas perpesanan internal dan notifikasi *nudge*.